

1. SPERMATOGENESIS

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Spermatogenesis allows the production of specialized reproductive cells, male gametes or spermatozoa, haploid, from diploid stem cells called spermatogonia. This process, continuous throughout the life of the man, takes place in the male genital tract, at the level of the seminiferous tubules of the testicle.

Spermatogenesis begins at puberty, reaches its peak around 20-30 years old, gradually slows down around forty but can continue into old age.

Spermatogenesis is a slow and continuous process taking about 74 days in humans; This is the duration of the sperm cycle.

1. REMINDERS ABOUT THE MALE GENITAL SYSTEM

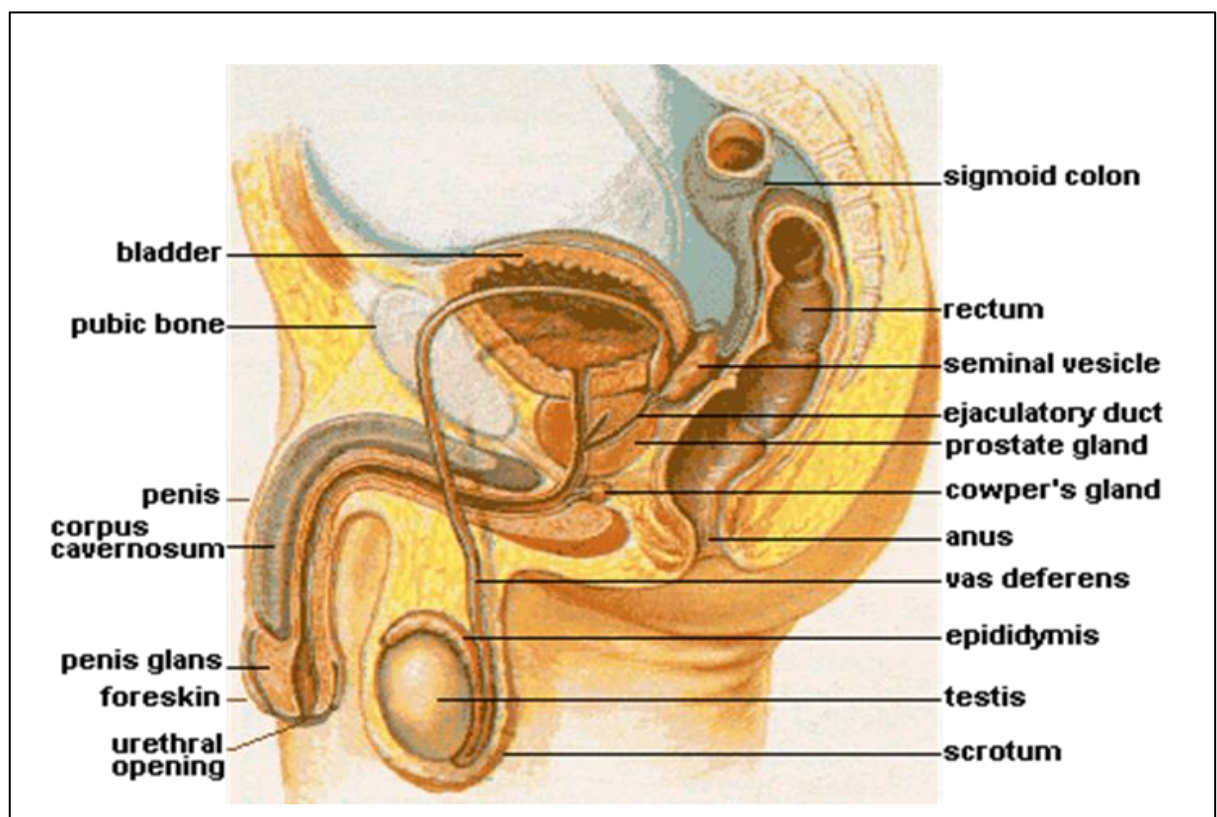


Figure 1. The male reproductive system

The male reproductive system includes the penis, scrotum, testes, epididymis, vas deferens, prostate, and seminal vesicles.

The testicle is an ovoid organ surrounded by the tunica albuginea, a fibrous connective tissue capsule with little elasticity, a thickening of which at the upper pole forms the Highmore body. Thin connective tissue septa delimit testicular lobules, numbering 200 to 300 per testicle. Each lobule contains a bundle of seminiferous tubules (1 to 4 per lobule). On the Highmore body side, the seminiferous tubules of each lobule merge into a straight tube 1 mm long. The straight tubes communicate with a network of canals running through the Highmore body, the rete testis.

The epididymis covers the testicle and has three parts of decreasing thickness from the upper pole: the head, the body and the tail. It contains two types of ducts: the efferent cones, located in the head of the epididymis, and formed by a spiral tube emanating from the rete testis; the epididymal duct, where the efferent cones converge, occupies the body and the tail of the epididymis, and extends outside the epididymis by the vas deferens.

The seminiferous tubule in cross-section, appears to consist of:

- a thin external connective membrane, called the proper membrane, containing contractile myoid cells and separated from the epithelium by a basement membrane;
- a so-called germinal epithelium, where two types of cells are visible, Sertoli cells and germline cells.

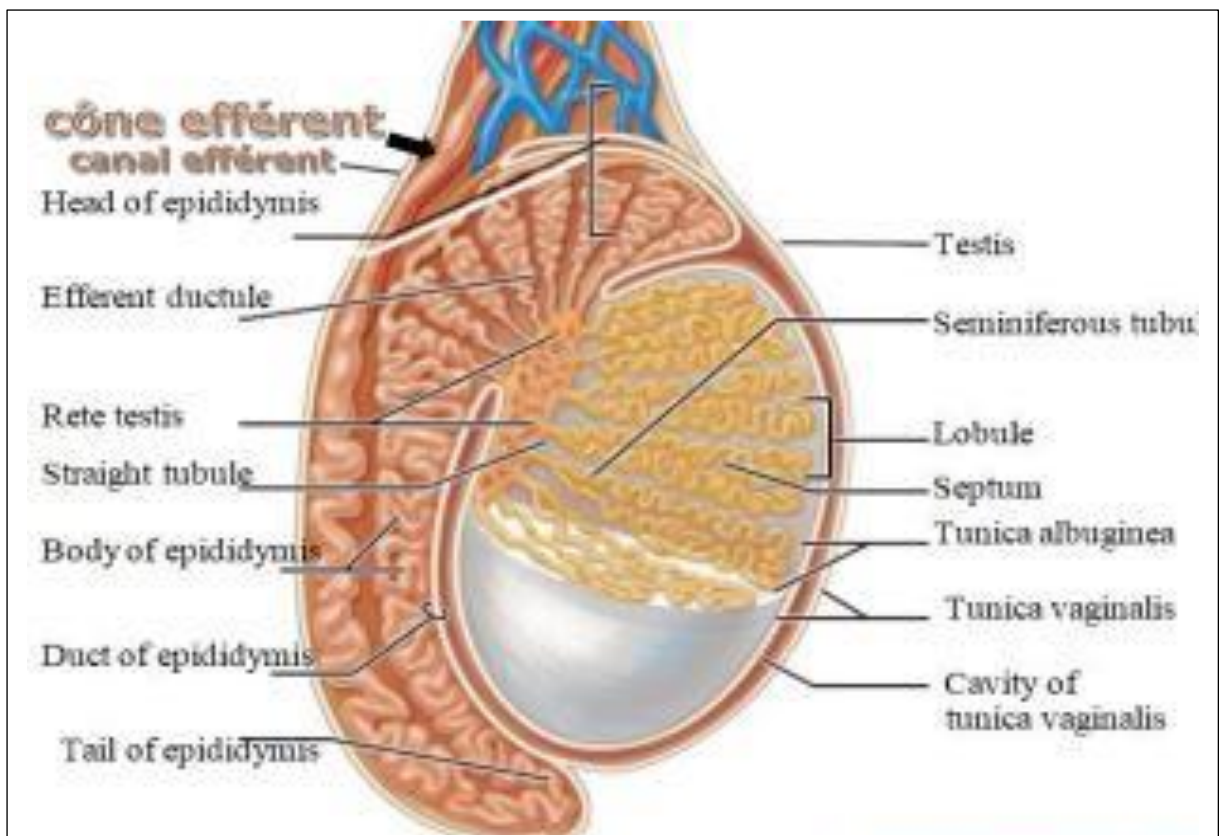


Figure 2. Structure of the testis

Sertoli cells are large, roughly pyramidal cells that occupy the entire thickness of the epithelium. They have multiple processes and are connected to each other by tight junctions located at the ends of their lateral processes.

Germline cells are located in the basal compartment, against the membrane proper, between the Sertoli cells, with which they are connected by various junction systems. Arranged in more or less regular layers, they represent, from the periphery towards the center of the tube

(adluminal compartment), the successive stages of spermatogenesis: spermatogonia, primary spermatocytes, secondary spermatocytes, spermatids and spermatozoa.

In the loose connective tissue surrounding the seminiferous tubules are islands of interstitial cells, or Leydig cells, arranged in cords around capillaries. They are responsible for the secretion of androgens (testosterone).

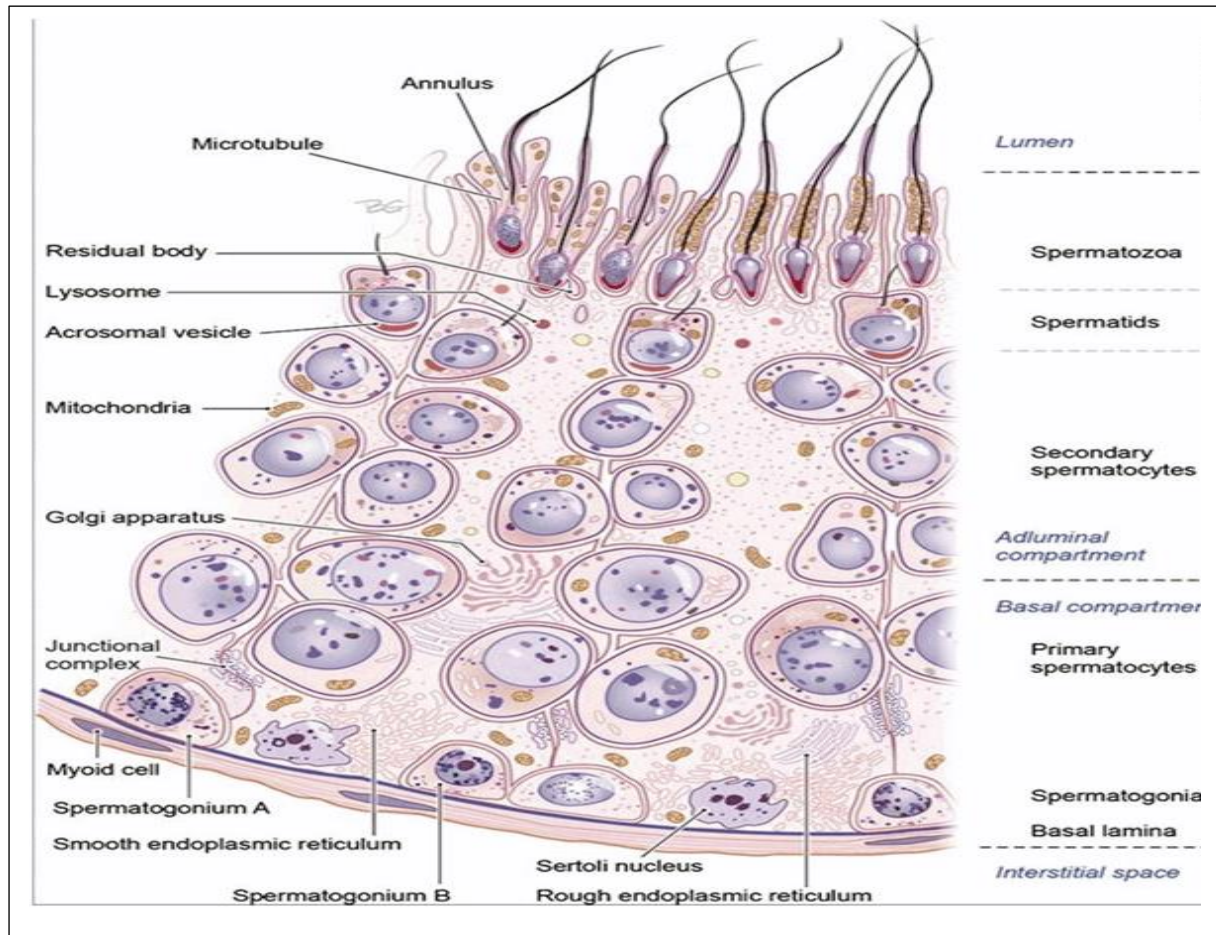


Figure 3. Spermatogenesis in the seminiferous tubules

2. STRUCTURE OF SPERM

The spermatozoon is a motile, highly differentiated cell, whose organization is comparable in most mammals.

Morphology

The spermatozoon is about 60 μm long. In light microscopy, we distinguish:

- **The head**, elongated and flattened (4 to 5 μm long and 2 μm thick). It contains the acrosome and the nucleus. -

- **The pass**, narrowed and short (1 μm by 1 μm), classically corresponding to the space between the 2 centrioles. -

- **The flagellum** including: the intermediate piece, the main piece and the end piece.

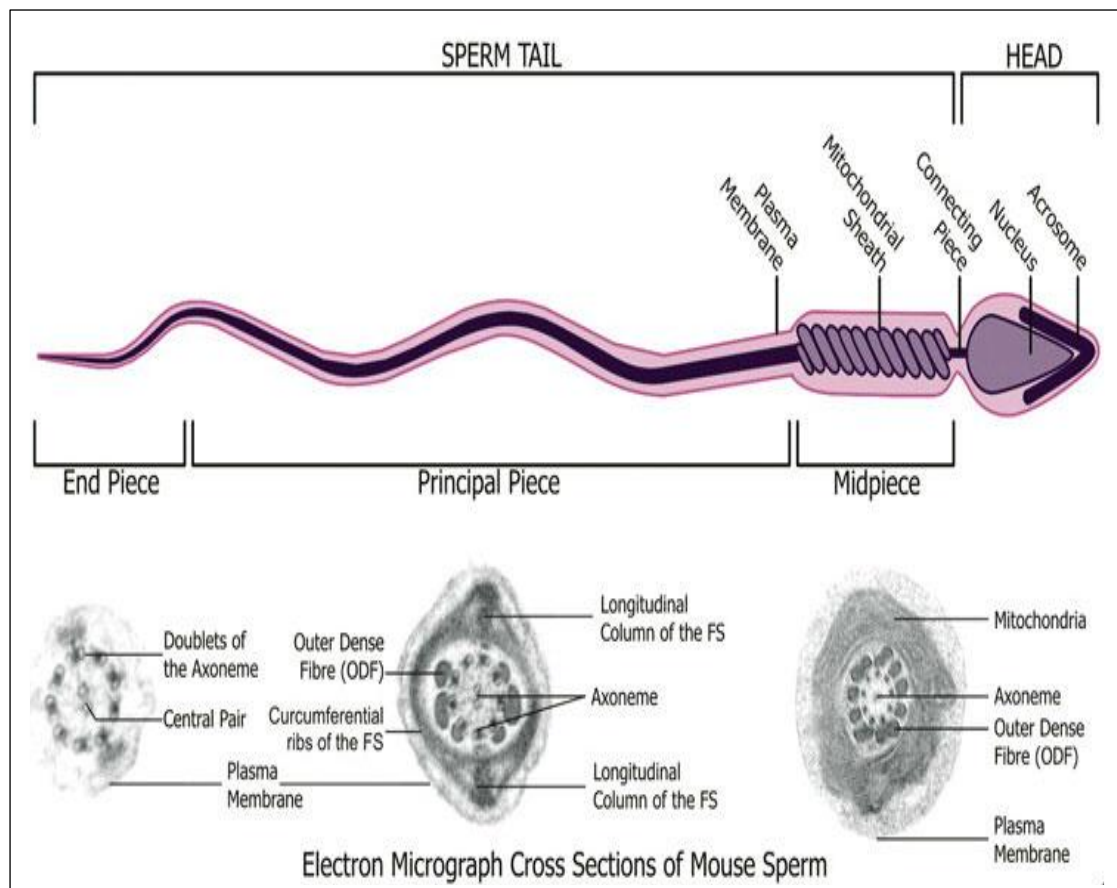


Plate 1. Structure and Ultra structure of human sperm

3. ULTRA STRUCTURE OF THE SPERM

- **The head:** It is oval and flattened. At the boundary between the anterior 2/3 and the posterior 1/3, a groove, the nuclear ring, corresponds to the posterior edge of the acrosome.
- **The core:** It is capped by the acrosome and occupies most of the head. Its chromatin is highly condensed, practically homogeneous, without nucleolus. The posterior pole of the nucleus has a transversely oriented depression, the implantation pit.
- **The acrosome:** It is flattened and covers the anterior 2/3 of the nucleus. It is a vesicle with homogeneous contents, limited by a membrane. It comprises 2 segments, with different appearance and physiological significance:
 - The main segment, in front, covers the anterior 1/3 of the nucleus and is 70 nm thick. It contains hyaluronidase. Its internal membrane is separated from the nuclear envelope by the subacrosomal space, 20 nm thick.
 - The equatorial segment, at the rear, covers the middle 1/3 of the nucleus. Its thickness is 25 nm. It is in this segment that acrosin is mainly located. The inner membrane is approximately 40 nm from the nuclear envelope.
- **The cytoplasm:** It is very reduced and constitutes a blade which surrounds the posterior third of the nucleus, behind the acrosome.
- **The pass:** or neck or collar or connective piece, it is the junction zone between the head and

the flagellum. It is a very complex region which contains the centriolar apparatus, the capitellum, and part of the segmented columns.

Starting from the core we can describe:

- **The proximal centriole:** It is located below the implantation fossa. Its axis is almost parallel to the posterior surface of the nucleus and makes an angle of 80° with that of the axoneme.
- **The 9 Dense Fibers:** They are made up of cytoskeletal proteins. They line the inner face of the segmented columns and extend towards the rest of the flagellum.
- **The axial filamentous complex:** It is formed by elongation of the distal centriole and begins approximately halfway up the segmented columns. Located centrally, within the dense fibers, it has the usual structure of a ciliary axis: 9 peripheral doublets surrounding a central doublet.
- **Mitochondria:** outside the dense fibers, they constitute the beginning of the mitochondrial sleeve which continues into the intermediate piece of the flagellum.
- **The flagellum** consists of 3 parts.
 - **The intermediate piece:** It is the shortest and thickest part of the flagellum. It is limited, at its distal end, by a thickening of the flagellum membrane, the annulus (corresponding to the Jensen ring of light microscopy).
In cross-section, we observe, from the center:
 - The axoneme or axial filamentous complex which has the typical structure of the ciliary axoneme: 9 doublets of peripheral microtubules surround a central doublet.
 - The 9 dense fibers - The spiral mitochondrial sleeve. It surrounds the dense fibers and is made up of a succession of mitochondria arranged in a single spiral line that makes approximately 40 turns.
 - The plasma membrane.
 - **The main room:** Its structure is identical along its entire length with, from the center:
 - The axial filamentous complex which continues that of the intermediate piece.
 - The 9 dense fibers. They become thinner towards the distal end
 - The fibrous sheath. These are fibrillar proteins wound in a spiral. This sheath has two diametrically opposed thickenings: the longitudinal columns. As one approaches the end of the main piece, the longitudinal columns disappear and the thickness of the fibrous sheath decreases.
 - The plasma membrane of the flagellum.
 - **The terminal piece:** It has a simplified structure with:
 - The axial filamentous complex in the center.
 - The plasma membrane.

4. PHASES OF SPERMATOGENESIS

Spermatogenesis begins around the age of 13-14, at puberty, and continues until a very advanced age. It takes place inside the seminiferous tubules. Its duration is fixed and characteristic of a species.

4.1. Multiplication phase

Spermatogonia, diploid stem cells ($2n$ chromosomes and $2n$ DNA), multiply by mitosis. They are the most peripheral germ cells. They are located in the basal compartment of the tube, in contact with the basement membrane and the Sertoli cells.

- Spermatogonia A1 or Ad (dark) are the true stem cells.
- Spermatogonia A2 or Ap (pale) represent the first cell stage of spermatogenesis proper and have a lifespan of 18 days before dividing into 2 B spermatogonia.
- B spermatogonia divide after 9 days to give 2 spermatocytes I, the 2 spermatocytes I do not separate completely, cytoplasm bridges persist until the end of spermatogenesis allowing exchanges between all the cells deriving from the same gony and allowing synchronous evolution.

4.2. Growth phase

It concerns spermatocytes I and corresponds schematically to the interphase and the beginning of the prophase of the first meiotic division. The size of the cells increases by accumulation of synthesized material. When they reach their maximum size, $25\text{ }\mu\text{m}$, they are called auxocytes.

4.3. Maturation phase (meiosis)

It concerns spermatocytes I. It is essentially a nuclear maturation, this phase corresponds to meiosis. Prophase is long (about 16 days). At the end of prophase, spermatocytes I contain $2n$ chromosomes and $4n$ DNA.

- The first division of meiosis (reductional division) It occurs after 23 days. Prophase continues with metaphase, anaphase and telophase. This first division results in the formation of two second-order spermatocytes or spermatocytes II. Each spermatocyte II has half the number of chromosomes present in spermatocyte I.
- The second division of meiosis (equational division) It occurs 24 hours after the first. Each secondary spermatocyte gives rise to 2 haploid spermatids that will no longer divide. Spermatids (n chromosomes, n DNA) are small, round cells.

Meiosis

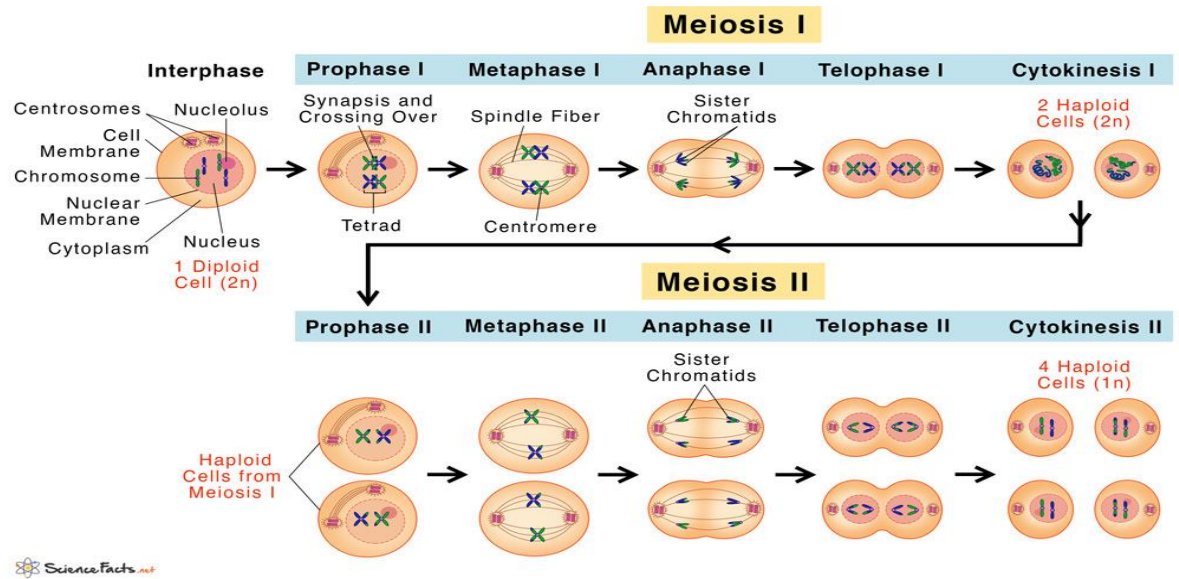


Figure 4. Simplified diagram of meiosis

4.4. Differentiation phase (spermiogenesis)

Spermiogenesis is specific to male gametogenesis. It corresponds to the transformation of each spermatid into a spermatozoon. This phase lasts 23 days and ends with the release of spermatozoa into the lumen of the seminiferous tubule, a phenomenon called spermiation. Spermiogenesis includes 5 fundamental phenomena which are practically simultaneous:

a. Formation of the acrosome from the Golgi

The Golgi vesicles develop, become pro-acrosomal granules and then fuse to form a single vesicle, the acrosomal (or acrosomal) vesicle, which spreads out and covers 2/3 of the nucleus. It is called an acrosome and its contents become homogeneous. The acrosome is a large lysosome that will be located at the anterior pole of the future spermatozoon. Its enzymatic content will be essential for crossing the oocyte envelopes during fertilization (hyaluronidase, acrosin).

b. Formation of the flagellum

The 2 centrioles migrate to the pole opposite the acrosome.

The microtubules of the distal centriole elongate and organize into an axoneme that elongates and emerges from the cell by pushing back the plasma membrane.

c. Nuclear elongation

Nuclear changes begin around the middle of spermiogenesis. The nucleus gradually elongates and becomes denser. The anterior two-thirds of the nucleus are covered by the acrosome. The posterior third is related to the cell's cytoplasm.

d. Formation of the mitochondrial sleeve

Mitochondria are arranged end to end in a spiral around the Initial portion of the axoneme, constituting the mitochondrial sleeve of the Intermediate piece. At its distal end, the sleeve is bounded by a dense ring described as the annulus.

e. Removal of cytoplasm

Most of the cytoplasm is phagocytosed by Sertoli cells. Some of it is detached, forming the residual bodies (Regaud's bodies). The rest will be eliminated in the final phase of maturation during migration into the epididymis.

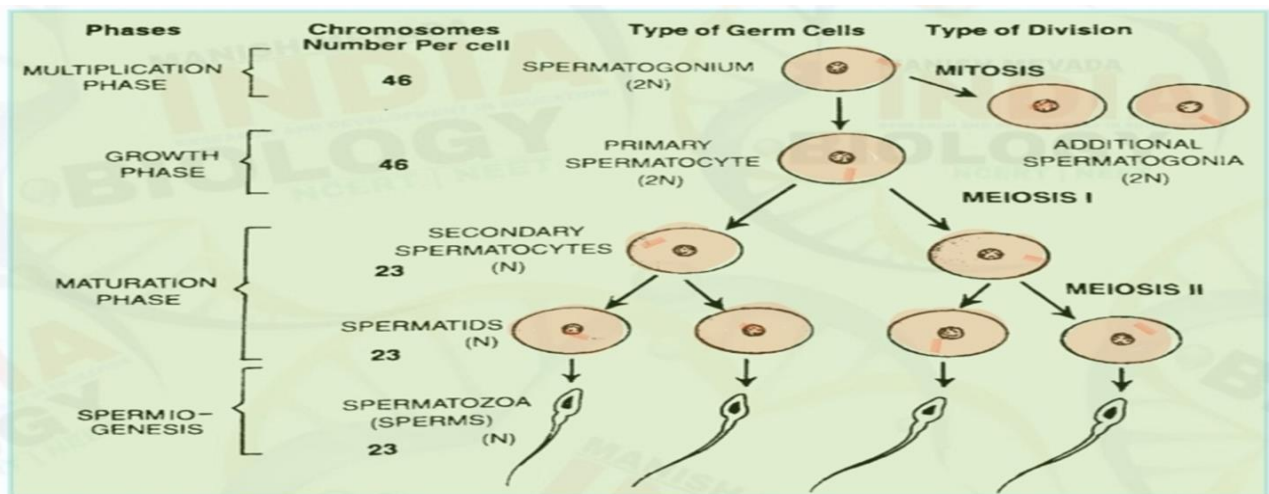


Plate 2. General diagram of spermatogenesis

5. EMISSION OF SPERM

Sperm are produced and formed in the testes and then released into the epididymis. They first pass from the rete testis to the head of the epididymis, then into the efferent cones, which continue through the ductus epididymis. The gametes are modified throughout their journey through the genital tract.

- In the epididymis, under the action of androgens, spermatozoa acquire their mobility.
- It is also in the epididymis that the spermatozoa are disabled: they then temporarily lose their ability to fuse with the other membranes.
- Sperm can be stored in the epididymis and survive there for about three weeks.
- The sperm then pass into the vas deferens, which is extended by the ejaculatory duct passing through the prostate and which joins the urinary canals. They then take the path to the urethra.

During their journey, the spermatozoa are mixed with liquids produced by the sexual glands (prostate and seminal vesicles); the mixture of these liquids gives sperm.

- The composition of the liquid phase of sperm can be detailed as follows: 5% epididymal secretions, 65% secretions from the seminal vesicles, 30% prostatic secretions

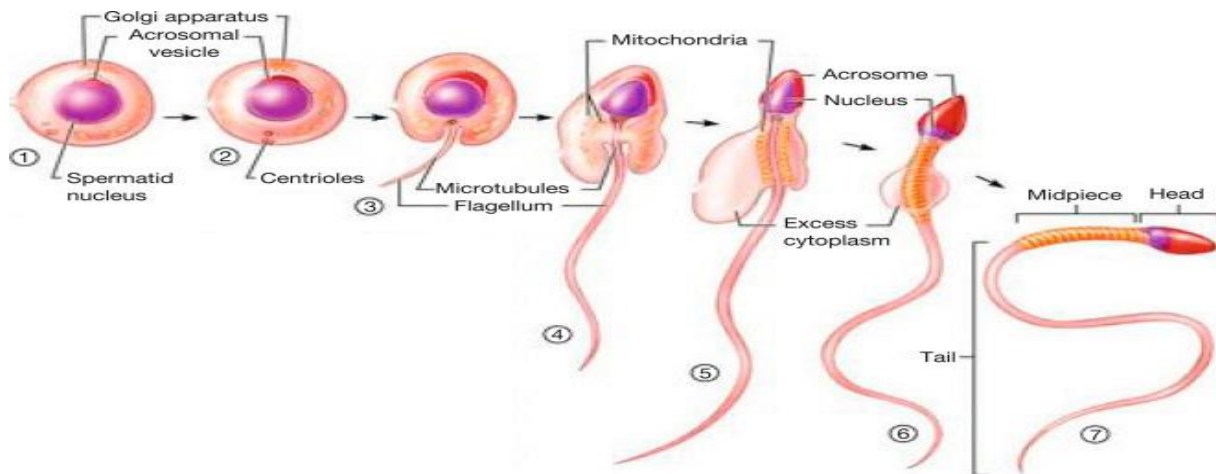


Figure 5. Stages of spermiogenesis

6. PHYSIOLOGICAL PROPERTIES OF SPERM

Motility

The flagellum's movements are undulatory and start at the proximal part of the flagellum. The spermatozoon moves in a sinusoidal pattern, with a variable velocity depending on the viscosity of the medium. The movement requires the hydrolysis of ATP.

Fertility

The spermatozoon only becomes fertilizing after capacitation, which is characterized by: - The loss of the glycoprotein coat of the spermatozoon's plasma membrane during passage through the female genital tract. - An increase in velocity. Vitality Seminal fluid is a poor survival environment for gametes. Only 20% of spermatozoa survive 24 hours at 37°C. On the other hand, in the female genital tract, survival can reach 2 or 3 days. After freezing in liquid nitrogen, the vitality and fertilization of some of the spermatozoa are preserved.

Metabolic activity Sperm have no nuclear activity. They do not synthesize DNA, RNA, or proteins. However, respiratory activity is important because flagellar movement requires a large supply of ATP.

7. FACTORS INFLUENCED SPERMATOGENESIS

7.1. Hormonal Control

Testicular function is established through the pulsatile production of GnRH (Gonadotropin Releasing Hormone) by neurons in the hypothalamus. GnRH causes the pituitary to secrete two hormones, FSH and LH. In the testicle, these hormones have the following actions:

:

- FSH enables the development of Sertoli cells and spermatogenesis (exocrine function of the testicle: excretion of spermatozoa). FSH binds to membrane receptors on Sertoli cells and plays a triple role:
 - it activates spermatogenesis through the sertolien cytoplasm;

- it stimulates the formation of ABP (Androgen Binding Protein);
- it causes the secretion of inhibin, a hormone exerting negative feedback on the secretion of FSH, either on hypothalamic neurons by reducing the secretion of GnRH, or directly on pituitary gonadotropic cells.

- LH ensures the multiplication of Leydig cells and the secretion of testosterone (endocrine function of the testicle):
 - most of the testosterone enters the sertolien cytoplasm where it binds to ABP to condition the development of the seminal epithelium and the proper functioning of the genital tract (seminal fluid).
 - free testosterone passes into the blood and exerts two actions: a positive action on the genital tract and the accessory glands and a negative feedback on the secretion of LH, either indirectly on the hypothalamic neurons or directly on the pituitary gonadotropic cells.

7.2. External factors

Temperature

Spermatogenesis can only take place at 34 or 35°C. This is the normal temperature of the testicles outside the body.

Deficiencies

NutritionalThe normal course of spermatogenesis requires an adequate quantitative and qualitative intake of proteins, particularly arginine. Protein malnutrition leads to alterations in spermatogenesis that are all the more serious the younger the adult. Vitamin A deficiency causes degenerative lesions of the spermatozoa. Vitamin E deficiency is accompanied by damage to the spermatids and spermatozoa.

Ionizing radiation

Spermatogonia multiply actively and are very radiosensitive. Chemical toxins Many chemicals can be toxic (pesticides, organochlorines).

Drogs

neuroleptics, steroid hormones, anti-inflammatories, chemotherapies.

Various exogenous factors

Alterations in general health, tobacco, alcohol, cannabis are known to alter spermatogenesis.

